Listing 7.21. A sample `push()` with helping for a lock-free queue

```
template<typename T>
class lock_free_queue
{
private:
    void set_new_tail(counted_node_ptr &old_tail,                    <-1
                      counted_node_ptr const &new_tail)
    {
        node* const current_tail_ptr=old_tail.ptr;
        while(!tail.compare_exchange_weak(old_tail,new_tail) &&      <-2
              old_tail.ptr==current_tail_ptr);
        if(old_tail.ptr==current_tail_ptr)                           <-3
            free_external_counter(old_tail);                         <-4
        else
            current_tail_ptr->release_ref();                         <-5
    }
public:
    void push(T new_value)
    {
        std::unique_ptr<T> new_data(new T(new_value));
        counted_node_ptr new_next;
        new_next.ptr=new node;
        new_next.external_count=1;
        counted_node_ptr old_tail=tail.load();

        for(;;)
        {
            increase_external_count(tail,old_tail);
```